

Multiple Choice

1. **Answer:** C

Options: A) biomass of all producers; B) biomass of all consumers; C) energy available to heterotrophs; D) energy consumed by decomposers; E) total solar energy converted to chemical energy by producers

Note: Net primary productivity is crucial in ecosystems because it quantifies the energy produced by autotrophs through photosynthesis, which forms the foundation of the food web and provides the energy necessary for heterotrophs to survive.

2. **Answer:** A

Options: A) The stomach has a high permeability to hydrogen ions; B) The presence of beneficial bacteria that neutralize stomach acid; C) The stomach lining is replaced every month; D) The stomach lining secretes digestive enzymes only when food is present; E) The production of bile in the stomach neutralizes the digestive acids

Note: The correct answer is based on the fact that the stomach's epithelial cells secrete a thick layer of mucus that protects the gastric lining from being eroded by its own acidic secretions, rather than having high permeability to hydrogen ions, which is a misleading statement.

3. **Answer:** A

Options: A) mistakes in translation of structural genes.; B) mistakes in DNA replication.; C) translocations and mistakes in meiosis.; D) recombination at fertilization.

Note: Mistakes in translation of structural genes do not contribute to genetic variation because they occur at the level of gene expression rather than affecting the underlying DNA sequence, which is the primary source of heritable variation for evolution.

4. **Answer:** A

Options: A) that humans evolved from apes.; B) that evolution occurs.; C) that evolution occurs through a process of gradual change.; D) that the Earth is older than a few thousand years.; E) a mechanism for how evolution occurs.

Note: The correct answer is based on Darwin's theory of evolution, which posits that humans and apes share a common ancestor, leading to the misconception that he proposed humans evolved directly from apes, while he actually emphasized the broader concept of evolution through natural selection.

5. **Answer:** A

Options: A) Compare gene frequencies and genotype frequencies of two generations.; B) Evaluate the population's response to environmental stress; C) Assess the mutation rates of the population's genes; D) Compare the population size over generations; E) Calculate the ratio of males to females in the population

Note: Comparing gene frequencies and genotype frequencies of two generations allows researchers to assess whether changes in allele distribution occur over time, which

indicates whether the population is in genetic equilibrium or undergoing evolutionary change.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: If messenger RNA molecules were not destroyed, their prolonged presence in the cell would allow for continuous translation by ribosomes, resulting in an unregulated and excessive production of proteins.

7. **Answer:** True

Options: A) True; B) False

Note: Autoradiography is a technique that uses radioactive isotopes to trace the incorporation of labeled amino acids into proteins, allowing visualization of protein synthesis and localization within cells.

8. **Answer:** False

Options: A) True; B) False

Note: Normal somatic cells of a human male contain one active X chromosome and one Y chromosome, as males are typically XY and have only one X chromosome.

9. **Answer:** False

Options: A) True; B) False

Note: Viruses cannot reproduce independently because they lack the necessary cellular machinery and metabolic processes, requiring a host cell to replicate and produce new viral particles.

10. **Answer:** True

Options: A) True; B) False

Note: The symplastic pathway for sucrose movement is true because it involves the interconnected plasmodesmata that allow sucrose to move through the cytoplasm of cells, including bundle sheath cells, phloem parenchyma, companion cells, and sieve tubes, facilitating efficient transport in the phloem.

Long Form Response

11. **Answer:** The first mammals emerged during the Carboniferous Period of the Paleozoic Era, evolving from synapsid ancestors. These early mammals were small, nocturnal creatures that adapted to a life in the shadows of dinosaurs, which would dominate the Mesozoic. In terms of reproductive strategies, monotremes, such as the platypus and echidna, lay eggs, showcasing a primitive reproductive adaptation that retains characteristics of their reptilian ancestors. In contrast, marsupials, like kangaroos and koalas, give birth to relatively underdeveloped young that continue to develop in a pouch, allowing for a more efficient allocation of maternal resources and a greater chance of

survival in a variety of environments. These differences highlight the evolutionary adaptations that have allowed both groups to thrive in diverse ecological niches.

Note: The answer correctly identifies the evolutionary origin of the first mammals during the Carboniferous Period, highlights their nocturnal adaptations, and distinguishes between the reproductive strategies of monotremes and marsupials by explaining the egg-laying characteristic of monotremes versus the live birth and pouch development of marsupials.

12. **Answer:** Classifying tissues in higher plants presents significant challenges due to their complex evolutionary heritage, which has resulted in a diverse array of tissue types that serve various functions. For instance, the differentiation between vascular tissues, such as xylem and phloem, and ground tissues can be difficult because these tissues have evolved to adapt to the specific environmental pressures faced by different plant lineages. Additionally, the presence of meristematic tissues, which can give rise to various cell types, complicates the classification further. Understanding plant biology necessitates recognizing these evolutionary contexts, as they influence not only the structure and function of tissues but also the overall adaptability and resilience of plant species in varying habitats. Therefore, a thorough comprehension of plant tissue classification requires an appreciation of both the functional roles of these tissues and their evolutionary pathways.

Note: correct because it emphasizes the complexity of plant tissue classification arising from evolutionary adaptations, which leads to diverse tissue types with specialized functions, complicating the distinction between categories like vascular and ground tissues.